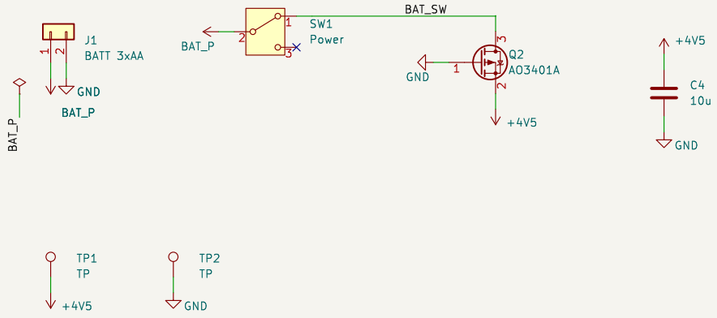
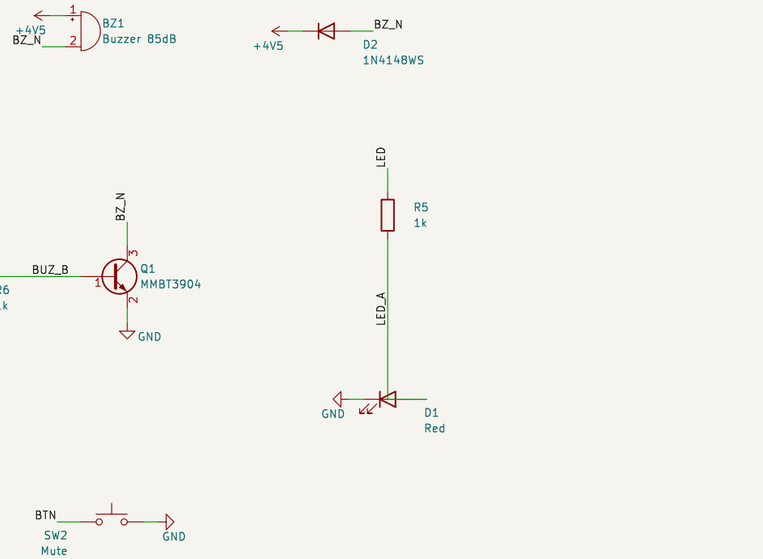
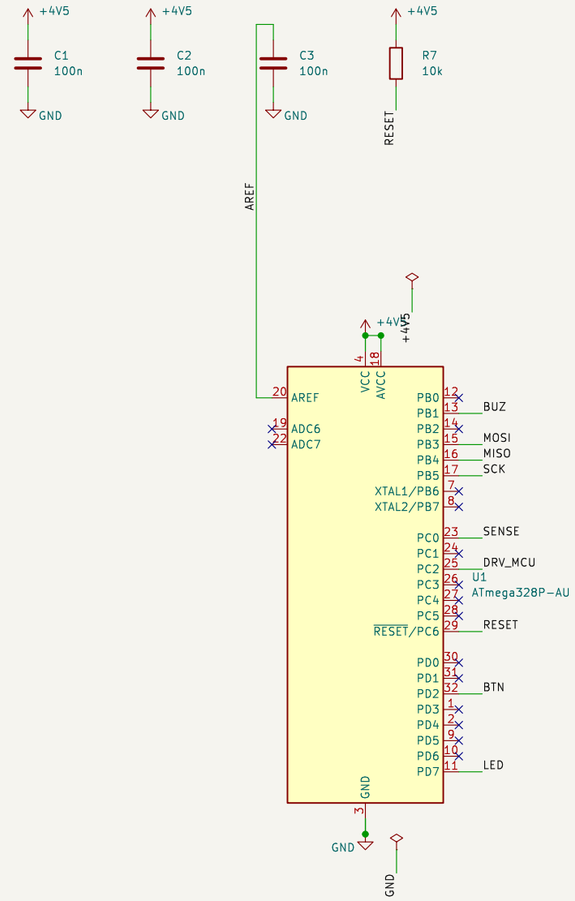
POWER INPUT: 3xAA, switch, reverse-polarity PMOS



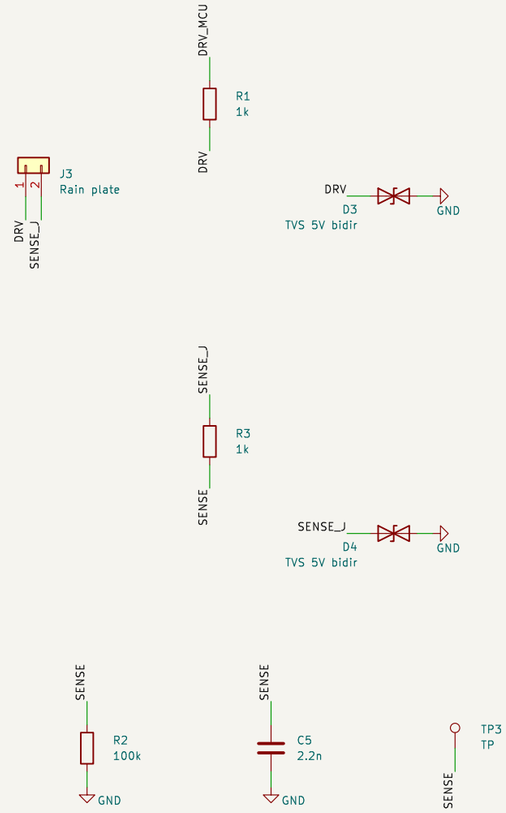
ALARM: buzzer driver, LED, mute



MCU: ATmega328P-AU, internal RC 8 MHz



RAIN PLATE INTERFACE: pulsed drive, series R + TVS



PROGRAMMING

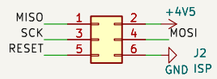


Plate is driven only while measuring (DRV high ~2–5 ms, then low): no DC across the wet plate, far less electrolytic corrosion.

SENSE node: plate R + 1k vs 100k pull-down, 2.2 nF filter. Cable lines have 1k series R and bidirectional TVS to GND (lightning/ESD).

3xAA 4.5 V; 8 MHz is within spec down to 2.7 V; fuses lfuse 0xE2 hfuse 0xD9 efuse 0xFD (BOD 2.7 V).

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /
File: rain_alert_main.kicad_sch

Title: Rain Alert Bell — indoor unit

Size: A3	Date: 2026-11-01	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1